

4	(a) e.m.f. = chemical energy to electrical energy p.d. = electrical energy to thermal energy idea of per unit charge	M1 M1 A1	[3]
	(b) $E = I(R + r)$ or $I = E / (R + r)$ (any subject)	B1	[1]
	(c) (i) $E = 5.8\text{ V}$	B1	[1]
	(ii) evidence of gradient calculation or calculation with values from graph e.g. $5.8 = 4 + 1.0 \times r$ $r = 1.8\ \Omega$	C1 A1	[2]
	(d) (i) $P = VI$ $P = 2.9 \times 1.6 = 4.6\ (4.64)\text{ W}$	C1 A1	[2]
	(ii) power from battery = $1.6 \times 5.8 = 9.28$ or efficiency = VI / EI efficiency = $(4.64 / 9.28) \times 100 = 50\%$ or $(2.9 / 5.8) \times 100 = 50\%$	C1 A1	[2]
5	(a) travel through a vacuum / free space	B1	[1]